

## **INA219 – Current, Voltage & Power Sensor (I<sup>2</sup>C based)**

### ♦ **What is INA219?**

INA219 is a **high-side current and power monitoring IC**.

It measures:

- **Bus Voltage (V)**
- **Current (I)**
- **Power (P)**

👉 Unlike basic current sensors, it measures **without disturbing the ground line**.

This breakout board will solve all your power-monitoring problems. Instead of struggling with two multimeters, you can just use the handy INA219 chip on this breakout to both measure both the high side voltage and DC current draw over I<sup>2</sup>C with  $\pm 1\%$  precision.

Can detect short circuits or overcurrent easily

### ♦ **Internal Principle (VERY IMPORTANT)**

INA219 uses:

- **Precision shunt resistor**
- **Differential amplifier**
- **ADC**
- **I<sup>2</sup>C digital interface**

📌 It measures **voltage drop across the shunt resistor** and calculates current using Ohm's Law.

| <b>Pin</b> | <b>Function</b>         |
|------------|-------------------------|
| VCC        | Power (3.3V or 5V)      |
| GND        | Ground                  |
| SDA        | I <sup>2</sup> C Data   |
| SCL        | I <sup>2</sup> C Clock  |
| VIN+       | Power input from supply |
| VIN-       | Power output to load    |

$$I = \frac{V_{shunt}}{R_{shunt}}$$

$$P = V_{bus} \times I$$

### Typical Connection (Arduino Example)

- VCC → 5V
- GND → GND
- SDA → A4
- SCL → A5
- VIN+ → Battery / Supply +
- VIN- → Load +

### You need INA219 ONLY IF your tile must:

- Measure **current consumption**
- OR calculate **power usage**
- OR monitor **battery health**
- OR show **energy efficiency**
- OR justify “**smart / energy-aware tile**” in evaluation

If your tile **only senses environment / touch / pressure / moisture** →

👉 **INA219 is NOT compulsory**

**INA219 goes in SERIES with the POWER LINE, not signal line**

### Correct order:

Power Source (+)



INA219 (VIN+)



INA219 (VIN-)



Tile Electronics (ESP / Arduino / Sensors)

📌 It sits **between battery/supply and entire tile circuit**

# HOW MANY INA219 DO YOU NEED?

## Case A: ONE TILE, ONE POWER SOURCE

### ✓ Only 1 INA219

- Measures **total tile power**
- Clean, simple, enough for project

## Case B: Multiple Tiles, Each Tile Independent

### ✓ 1 INA219 per tile

- Each tile reports its own consumption
- Scalable & impressive

## Case C: Want sensor-wise consumption

- ✗ Overkill
- ✗ Confusing
- ✗ Not required for college projects

### 👉 FINAL ANSWER FOR YOU:

**1 INA219 per tile** is PERFECT.

## Power Routing (MOST IMPORTANT)

- Battery / Adapter + → **INA219 VIN+**
- **INA219 VIN-** → **Tile VCC rail**
- Ground is common

### INA219 Controller

VCC      3.3V / 5V

GND      GND

SDA      SDA

SCL      SCL

📌 No extra ADC needed. Fully digital.

## Correct physical placement:

- **Near power input**
- Before voltage regulators
- Away from high-noise motors (if any)

 Reason:

- Accurate current measurement
- Less noise
- Cleaner readings

<https://www.ti.com/lit/ds/symlink/ina219.pdf>

<https://learn.adafruit.com/adafruit-ina219-current-sensor-breakout/downloads>

## Key Features:

- **Wide Voltage Sensing Range:** Monitors bus voltages from 0V to 26V for both low and high-voltage applications.
- **Integrated Power Monitoring:** Reports voltage drop, current draw, and power consumption in real time.
- **I2C/SMBus Interface:** Seamless integration with microcontrollers via I2C or SMBus communication.
- **High Accuracy Measurements:** Voltage ( $\pm 0.4\%$ ), current ( $\pm 1\%$ ), and power readings for precise system diagnostics.
- **16 Programmable Addresses:** Enables monitoring of multiple modules on the same bus.
- **Programmable Calibration Registers:** Compensates for component tolerances to ensure accuracy.
- **Low Quiescent Current:** Operates efficiently with only 1mA standby current, perfect for low-power designs.
- **Alert Pin Functionality:** Triggers system interrupts based on configurable thresholds.
- **Built-in Gain Amplifier:** Supports eight programmable gain levels for sensitivity across a wide range of currents.
- **Filtering Options:** Adjustable filtering improves stability in noisy environments.

| Item                        | Quantity | Approx Cost (₹)  | Notes                       |
|-----------------------------|----------|------------------|-----------------------------|
| INA219 Sensor Module        | 1        | <b>180 – 250</b> | Depends on seller & quality |
| Jumper wires / PCB traces   | —        | <b>10 – 20</b>   | Usually already counted     |
| Extra PCB space             | —        | <b>0</b>         | Negligible                  |
| Power loss (shunt resistor) | —        | <b>0</b>         | Very low (milliwatts)       |

👉 **Total INA219 Integration Cost per Tile:**

**₹190 – ₹270**

| Sensor | What “pressure” actually means |
|--------|--------------------------------|
|--------|--------------------------------|

|       |                                                             |
|-------|-------------------------------------------------------------|
| HX711 | <b>Mechanical force / weight</b> (someone stepping on tile) |
|-------|-------------------------------------------------------------|

|        |                                                             |
|--------|-------------------------------------------------------------|
| BME280 | <b>Atmospheric pressure</b> (air pressure, weather-related) |
|--------|-------------------------------------------------------------|

👉 These are **completely different domains**.  
So **NO overlap** here.

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♦ **INA219 vs others**

- INA219 measures **electrical power**
- HX711 & BME280 measure **physical/environmental parameters**

👉 **ZERO overlap**

### What your tile currently does

- ✓ Detects **step / load / force**
- ✓ Measures **environmental conditions**
- ✓ Monitors **energy consumption**
- ✓ Supports **safety & efficiency analysis**
- ✓ Looks **interdisciplinary** (mechanical + environmental + electrical)

✗ **NO single sensor can replace HX711 + BME280 + INA219**

| Parameter        | Needs                          |
|------------------|--------------------------------|
| Mechanical force | Strain gauge + ADC             |
| Environment      | MEMS temp/humidity             |
| Electrical power | Shunt + ADC + I <sup>2</sup> C |

These are **physically different phenomena**.  
Trying to merge them = bad engineering.

| Sensor            | Cost (₹) |
|-------------------|----------|
| HX711 + Load Cell | 250–350  |
| BME280            | 300–450  |
| INA219            | 190–270  |

**Total:**

**₹750 – ₹1050**

For **3 completely different sensing domains**, this is actually **very cost-efficient**.